# Girls Build Bots — Brownie App Development

## Badge-aligned workshop guide for robotics-team-led outreach

**Level:** Brownies (Grades 2–3) · **Badge:** Coding for Good — App Development **Series:** Coding for Good (Coding Basics · Digital Game Design · App Development) **Format:** One 75–90 minute session · **Ratio:** 1 mentor per 3–4 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on free browser tools and craft supplies the team or troop can provide.

> **Troop Leaders:** This curriculum is designed to *align with* the Brownie Coding for Good badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

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## The through-line: coding *for good*

Every badge in this series ties coding to helping people. Brownies should leave believing that code isn't just for games — it's a tool they can use to make something better for someone else. Mentors: name this out loud. "We're not just coding — we're solving a problem for a real person."

**Vocabulary:** app, user, problem, design, prototype, screen, button.

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## App Development (75–90 minutes)

**Badge theme:** Exploring how apps solve problems for people, and designing an app that helps her community.

### Materials

- "App design" paper templates (blank phone outlines — 3 screens per Scout)
- Markers, sticky notes, scissors
- Optional: tablets to show 2–3 real helper-apps (weather, maps, a timer)

### Activity 1 — Apps Are Problem-Solvers (15 min)

Look at a few apps the Scouts know. For each, ask the key question: **whose problem does this solve?** A weather app helps you know what to wear; a maps app helps you not get lost. Apps aren't magic — someone noticed a problem and built a fix. This is the "for good" heart of the badge.

## Activity 2 — Find a Problem Worth Solving (15 min)

Brainstorm: what's a problem in their school, troop, home, or neighborhood? (Remembering to water class plants, finding lost library books, knowing when the bus comes.) No idea is too small. Each Scout picks **one** problem her app will solve.

## Activity 3 — Design Your App on Paper (40 min)

Using the phone-outline templates, Scouts design a **paper prototype** — the real way app designers start: - **Screen 1:** the welcome screen (app name + what it does) - **Screen 2:** the main screen (the buttons that solve the problem) - **Screen 3:** what happens when you tap the main button

Mentors circulate asking: "What happens when someone taps here?" Paper apps don't need to work — they need to show the *idea*.

## Activity 4 — "Click Test" & Share (10 min) + Wrap (5 min)

In pairs, one Scout "uses" the other's paper app by pointing at buttons while the designer swaps the screens by hand — a real technique called a click-through test. Closing circle: each Scout names her app and the problem it solves in one sentence.

> **Robot moment (optional):** Share for 60 seconds how the team decides what the robot's control app needs before they build it — same "who needs this, what does it do" thinking.

## Completion Checklist — App Development

By the end of the session, to earn the badge, each Scout can: - [ ] Explain that apps are built to solve problems for people - [ ] Name a real problem in her community an app could help with - [ ] Design a multi-screen paper prototype of her app - [ ] Show how her app works with a "click test" and describe who it helps

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# Mentor Guide (for the robotics team)

- **You don't need a robot for this.** It runs entirely on paper and markers. If you brought a robot, the "robot moment" box is a 60-second add-on — not the main event.
- **Never solve it for her.** Guide with questions ("what would help the person using this?").
- **Every idea is welcome.** No problem is too small for a first app; celebrate the noticing.
- **Keep it small.** Three screens, one problem. Rein in over-ambition gently.
- **Watch the clock.** Brownies fade after ~25 minutes on one thing; move to the next segment on schedule even if unfinished.
- **Name the "for good."** The whole badge is about solving a real person's problem — say whose problem, out loud, often.

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## 📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

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*Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Brownie Coding for Good badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.*